DAY 6 — Introduction to Machine Learning & Supervised Learning Concepts

Today we started the world of **Machine Learning**, which was a major milestone. ML is the ability of computers to learn from data without being explicitly programmed.

🔹 What is Machine Learning?

A method through which machines improve their performance by learning patterns from datasets.

Types of Machine Learning:

- **Supervised Learning**
- **Unsupervised Learning**
- **Reinforcement Learning**

Our main focus today was **Supervised Learning**.

🔸 1. Supervised Learning

Supervised learning uses labeled data.
Examples:

- Predicting stock prices
- Predicting house prices
- Classifying emails as spam / not spam

Algorithms learned:

- Linear Regression
- Logistic Regression
- Decision Tree
- Random Forest

🔸 2. Regression vs Classification

Regression

Predicts continuous values
(e.g., stock prices, temperature)

Classification

Predicts categories
(e.g., fraud detection, sentiment analysis)

🔸 3. Training & Testing Split

We used:

from sklearn.model_selection import train_test_split

The dataset was divided into:

- 80% training
- 20% testing

This ensures fairness and avoids overfitting.

🔸 4. Evaluation Metrics Learned

For regression:

- MAE (Mean Absolute Error)
- MSE (Mean Squared Error)
- RMSE (Root Mean Square Error)

For classification:

- Accuracy
- Precision
- Recall
- F1 Score

Understanding these metrics is important for comparing models.

🔸 5. Building First ML Model

We built a basic **Linear Regression Model**:

- Fit the model
- Predict values
- Evaluate metrics
- Plot predictions vs actuals

This hands-on practice made ML concepts extremely clear.

Key Takeaways from Day 6

- ML is all about learning from data
- Supervised learning uses labeled datasets
- Evaluation metrics help judge model performance
- Linear Regression is the simplest yet powerful algorithm

This day marked the beginning of the real machine learning phase of my training.